

HUG Smart Sticker: Enhancing Personalized Intelligent Medication Management for Community-Dwelling Older Adults with an AIoT Intervention

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Abstract

Medication and dietary supplement usage are prevalent in community-dwelling older adults. However, medication or dietary supplements non-adherence is a challenge, as are risks of misuse, abuse, and diversion (MAD). Due to the increased prevalence of progressive deficits in cognition among community-dwelling older adults, the ability to plan, organize, and execute medicine-management behaviors is further compromised, leading to an increased risk of unintentional non-adherence, medication errors, and preventable medication-related hospitalizations. HUG Smart Sticker is a personalized intelligent medication management system promoting informed and safe medication usage for community-dwelling older adults. This Artificial Intelligent of Things (AIoT) intervention records users' medication usage in real-time, reports the latest alerts or potential side effects, and reminds users to ensure the safe usage of medications. Our study highlights the significance of this AIoT-based medication management intervention in improving medication adherence, enhancing the medication adherence of older adults, and contributing to aging populations' healthy lifespans.

CCS Concepts

• **Human-centered computing** → **Empirical studies in interaction design**; *Accessibility systems and tools*; • **Social and professional topics** → **Seniors**; • **Applied computing** → *Consumer health*.

Keywords

Artificial Intelligent of Things (AIoT); Aging populations; Community-Dwelling Older Adults; Medication Management Intervention.

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1 Introduction

In 2020, the US population aged sixty-five and older numbered 55.8 million, with older adults representing 16.8 percent of the population [11, 12]. Medication adherence use among older adults especially for those with cognitive decline is a major challenge, as are risks of misuse, abuse, and diversion (MAD) of medication [28, 30, 44]. Cognitive impairment interferes with the ability of older adults to comply with their medication regimen. The overall increase in longevity globally and greater rates of cognitive impairment and utilization of medications that come with older age suggest that the already large population of older adults exhibiting medication non-adherence will increase [20, 41, 46, 53].

The average medication nonadherence rate is 24.8%, according to a 2004 study [14]. 89% of older adults report that they are currently taking at least one prescription medicine, and the number of medications tends to increase with age [1, 5]. The misuse and abuse of medications can have serious ramifications for older adults' well-being [10, 17]. It is estimated that medication nonadherence contributes to around 100,000 deaths per year in the United States [30]. Medication non-adherence results in an increased risk of MAD-related behavior [25]. MAD of medications is associated with increases in admissions to addiction treatment centers, emergency department visits, and a number of prescribed medication-related poisoning deaths.

Effective interventions are critically needed to promote the appropriate use of prescribed medications to optimize pain management and minimize risks of both intentional and unintentional MAD. The current lack of potent, feasible, implementable, and scalable programs for community-dwelling older adults and their primary caregivers leaves this population, their families, and their communities at heightened risk for errors and likely contributes

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to the continuation of the increased prevalence of adverse drug reactions and potentially avoidable hospitalizations [37].

This paper investigates a research question pertaining to medication management practices among community-dwelling older adults: Based on their personal experiences, **how can we increase medication adherence in community-dwelling older adults?** To address this research question, we designed an AIoT-based intervention-HUG Smart Sticker-to maximize adherence and minimize unintentional and intentional MAD behavior for community-dwelling older adults.

HUG Smart Sticker improves medication safety and adherence for community-dwelling older adults experiencing age-related cognitive decline or cognitive impairment. Through our design study, we aim to contribute to the field of aging-friendly intelligent medication management interventions, particularly promoting medications and dietary supplements' adherence, enhancing safety, and addressing the unique challenges faced by community-dwelling older adults. With its unique 3R features (Record, Report, Remind) and aging user-friendly interface, the HUG Smart Sticker offers a tailored approach to medication and supplement management, empowering older adults to take control of their health and well-being. Furthermore, HUG Smart Sticker provides a comprehensive and aging populations-centered solution that empowers older adults to navigate the complexities of medication management, leading to improved medication adherence and enhanced quality of life.

2 Related Work

2.1 Challenges of Medication Non-Adherence Among Community-Dwelling Older Adults

Community-dwelling older adults living with cognitive impairment warrant specific attention because they are particularly vulnerable to medication non-adherence [24, 33, 52]. This can be attributed to the increasing trend towards home-based care in lieu of residential care and the heightened intensity of medical therapy to treat chronic diseases, which together suggest that a greater number of community-dwelling older adults with cognitive impairment will need to manage increasingly more complex medication regimens at their homes and will likely need assistance from caregivers [34, 58]. Smith et al.'s systematic review reveals that cognitive impairment significantly impacts medication adherence, with adherence rates ranging from 10.7% to 38%, a cause for concern [49].

Medication non-adherence is both varied and complex, and previous research has described the ability to manage medication as a delicate balance between patient factors [39, 54], healthcare professional factors, and medication regimen-related factors [15, 19, 21, 31, 35]. Typical medication non-adherence reasons cited by patients for not taking their medications included forgetfulness (30%) [50], other priorities (16%), decision to omit doses (11%), lack of information (9%), and emotional factors (7%) [13, 42]. Gathered insight shows that a combination of strategies is more effective than any single intervention and identifies the need for multifaceted and individualized approaches. The lack of comprehensive solutions leaves providers unable to coordinate with each other and meet the complex needs of their patients [4].

Apart from improved patient quality of life, there are measurable economic benefits to improved medication adherence among older

adults with cognitive decline [7, 8, 40]. The downstream cost associated with medication non-adherence is staggering and can have far-reaching consequences, with one estimate suggesting that medication non-adherence results in as many as 100,000 preventable deaths and as much as \$100 billion in preventable medical costs per year in the United States [27, 30]. The lack of medication adherence disrupts the continuum of care and necessitates additional medical visits, interventions, and potentially more expensive treatments. On the positive side, studies have found that in certain disease states, adherence improves health outcomes and reduces annual prevention costs by 10-17% [9]. Despite the evaluation of many intervention strategies and medication adherence devices, there is no standardized approach to sustaining success [18, 51]. Several strategies have been used to support adherence, yet little research underpinning the use of these strategies exists, and few have sought to turn these approaches into routine practice [20].

2.2 AIoT-based Integrations in Medication Management Systems

AIoT system is a combination of Artificial Intelligence (AI) and the Internet of Things (IoT) that facilitates the connection of physical objects and equipment. This integration empowers these entities with perception, cognition, and communication capabilities. AIoT technologies enhance the intelligence of previously non-intelligent devices, ultimately connecting them to the Internet by leveraging embedded devices, internet protocols, sensor networks, communication protocols, and various applications [45].

Prior research has focused separately on AI and IoT integrations in medication management systems to improve efficiency and effectiveness. Studies on AI tools can be used to measure and increase medication adherence in patients with non-communicable diseases (NCDs) [2]. AI technology could also detect errors and improve accuracy in medication self-administration by analyzing the wireless signals in the patient's home [57]. The smart AI pharmacy system could automate routine tasks, provide personalized treatment plans, and reduce costs without relying on redundant manual processes [29, 32]. In conjunction, IoT devices create optimal medication environments by controlling factors like temperature and humidity [47]. The Smart Medicine Dispenser (SMD) IoT system ensures accurate dosages and timely dispensing [43, 48]. Integrating AI and IoT offers potential for enhanced medication management, including adherence monitoring, error detection, automation, personalization, and precise delivery [6, 38, 55].

AIoT systems, combining the power of AI algorithms and IoT techniques, have been used in multiple fields, such as smart homes [16] and healthcare [3, 23]. There is limited research on AIoT integrations in the medication management field, but the AIoT systems enable intelligent and autonomous decision-making, data analysis, and communication among interconnected devices which can build a community-dwelling older adults-centered medication management system and benefit aging populations [36].

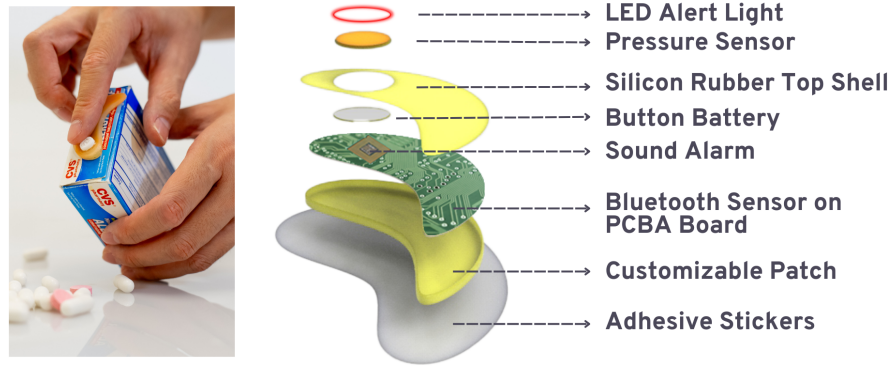


Figure 1: HUG Smart Sticker Prototype Look and Explosion Image of the Inner Functions.

3 Design Concept and Prototypes

3.1 Design and Develop the Older Adults-Friendly HUG Smart Sticker

This study aimed to develop an AIoT-based medication management intervention to 1) effectively manage community-dwelling older adults' medications or dietary supplements without altering their original behaviors [26]; 2) provide personalized 3R functions - Record, Remind, Report - to improve older adults' education on the medications that they are taking; and 3) predict users' medication adherence and behaviors to maximize the medication or dietary supplements' effectiveness. HUG Smart Sticker is designed as a two-arm intervention composed of an intelligent sticker and alerts for promoting informed and safe medication and dietary supplement usage. It is intended to optimize the health of the average older adult, as medications and dietary supplements are commonly used by the public in hopes of maintaining or improving health outcomes.

To achieve precise monitoring of users' medication or dietary supplements usage, we conducted extensive testing on various sensor types, including accelerometer, proximity sensor, and button sensor. Through multiple rounds of rigorous testing, we determined that the pressure sensor exhibited the highest level of accuracy and provided the best user experience among all the sensors evaluated. We also tested different types of visual, audio, and Bluetooth/WiFi data transmission sensor in order to form seamless communication and integration between sensors.

Each HUG Smart Sticker hardware module features an LED button light, a pressure sensor, a Bluetooth transmitter, an audio signal generator, batteries, and a silicone casing. The LED button light serves as a visual indicator, lighting up to remind users when it's time to take their medication. The pressure sensor is integral to tracking medication usage, registering the pressure change when a medication container is accessed and the sensor is pressed by the user. The Bluetooth transmitter is responsible for sending this information to the paired software application, allowing users to monitor and track their medication usage. The audio signal generator aids in this process, generating sound alerts based on users' medication schemes. The sticker's power source is an energy-efficient battery that ensures long-lasting operation. The adhesive silicone casing

encloses all these components, providing a robust and durable shell that can withstand daily wear and tear. The casing also allows users to customize the sticker easily, cutting and adjusting it to fit and attach to a specific medication container or any other packaging for tracking purposes.

One of the notable features of the HUG Smart Sticker is its customization capability. Users have the flexibility to easily modify and tailor the sticker to fit their specific medication containers or any other preferred location for tracking purposes. The sticker can be conveniently cut and adjusted according to individual preferences, ensuring seamless integration into the user's daily routine and enhancing user-friendliness. This customization feature adds versatility and adaptability to the HUG Smart Sticker, making it a practical and adaptable solution for medication management.

3.2 User flow of HUG Smart Sticker: Combining the Hardware and Software Modules

HUG Smart Sticker generates information about the 3R functions (Record, Remind, and Report) based on the users' medication intake data. To use the HUG Smart Sticker software alert system, users or caregivers (under permission) first match each hardware sticker with the corresponding medication's QR code. After confirming the medications or dietary supplements' doses and durations, users can check the medication usage report for each medication container. The intelligent HUG Smart Sticker can then be attached to the desired medication container. The wireless pressure sensor in the HUG Smart Sticker is able to detect the user's frequency of medication usage based on the time duration and amount of pressure applied to the container. For example, each time the user takes a medication, the label will record this one-time usage based on the pressure it senses from the user's grip on the HUG Smart Sticker button. The pressure sensor will then transmit this medication-tracking data via Bluetooth or WiFi to the user's end through a notification that will enable the user to monitor their supplement usage. Overall, the HUG Smart Sticker is intended to provide records of daily medication usage via the intelligent label and alerts on adherence to the medication regimen.

Building upon users' data, the HUG Smart Sticker will utilize AI algorithms to anticipate and predict users' future medication adherence behavior and patterns. This data-driven predictive feature will

contribute to a more personalized and patient-centric approach, ultimately promoting community-dwelling older adults' better medication adherence. It offers *intelligent* alerts and reminders to assist users in effectively managing their medication usage. Moreover, these predicted behavior patterns can be shared with clinical professionals and healthcare institutes/pharmaceutical companies, enabling them to gain insights into medication adherence. For instance, in cases where older adult users frequently exhibit medication overdosing, clinicians may opt to explore alternative treatment options to optimize the health outcomes and overall well-being.

The combination of hardware and software modules forms a comprehensive AIoT intervention (Figure 2), which offers community-dwelling older adults a convenient solution for real-time medication tracking and understanding of their medication usage. Users can initiate the medication intake and activate the HUG Smart Sticker's pressor sensor. The HUG Smart Sticker incorporates three types of alerts: hardware sticker alerts, software alerts, and alerts for caregivers. Subsequently, the data points are transmitted to the database, forming behavior patterns and enabling the HUG Smart Sticker to enhance its intelligence in delivering alerts. Following AI integration and analysis, the database generates personalized medication adherence reports including non-adherence rates, derived from data on misdoses, overdoses, prescribed dosages, refill status, etc. These reports, contingent on user consent, can be communicated to caregivers and healthcare institutes/pharmaceutical companies. The AIoT intervention establishes a robust foundation for medication management centered around the needs of older adults.

In the broader context, the medication usage data of each user can provide insights for clinicians to comprehend the performance of the prescribed medications. Since medication effectiveness is not solely reliant on the medication itself, it is also based on the consistent and regulated intake that maximizes its therapeutic benefits. Therefore, clinicians can gain a deeper understanding of medication performance and make informed decisions to optimize patient outcomes.

Furthermore, the availability of such data can have long-term benefits for pharmacies and pharmaceutical companies. It allows them to explore the effectiveness of medications and supplements from a user-centered perspective. By analyzing adherence patterns and correlating them with health outcomes, pharmaceutical companies can gain insights into the real-world effectiveness of their products. This user-centered approach promotes the development of tailored interventions, improves medication designs, and targets healthcare systems that better cater to the needs of older adults. In the long term, the utilization of HUG Smart Sticker-generated data not only benefits community-dwelling older adults by enhancing their awareness of medication usage, but also facilitates research and innovation in the pharmaceutical industry, which ultimately leads to improved healthcare outcomes for this population.

4 Study Process and Result

we recruited 7 community-dwelling older adult participants (Mean = 72.14 years, SD = 2.48 years) to understand their demands and challenges in managing the medications or supplements. (Table 1 & 2).

4.0.1 The lack of knowledge about medications or dietary supplements. The problem of inadequate knowledge about medications and dietary supplements is a prevalent issue experienced by community-dwelling older adults. Rather than inquiring specific medical recommendation from clinical professionals, society or surrounding environment has negative influence on getting accurate knowledge about medications or dietary supplements, which highlights the impact of a knowledge deficit: *"I have limited knowledge about my medications. I have been advised by my friends to take certain medications occasionally, such as calcium, Vitamin D, and other supplements that are relevant to my age or a general multivitamin."* (P1) In the absence of clinical suggestions or advice, the independent decision-making regarding medication intake poses potential risks for older adults: *"I've been taking a daily multivitamin called 'Centrum for Women' for years now. I must admit, my decision to start taking it was more out of routine and societal influence than a specific medical recommendation."* (P6)

In addition, insufficient understanding regarding the usage, benefits, and potential risks associated with these substances poses a considerable challenge, which may generate false belief: *"I don't have a lot of knowledge about certain dietary supplements, and I have reservations about taking them. I believe in getting my dietary supplements from my food."* (P3) P3 voiced a preference for obtaining essential nutrients from their food rather than resorting to medications, which mistakenly dismissing the role of medications in meeting their nutritional needs.

4.0.2 Limited tracking of medications and dietary supplements in non-traditional formats. 3 participants raised concerns regarding the types of containers used for medications or supplements. They noted that not all medications or dietary supplements are in solid form, with some being in powder, liquid, cream, or bag formats. *"But some medication cream or powder is hard to take care of or put in my medication pillbox. A smart sticker on the original package would definitely help a lot, I believe."* (P2) Tracking these non-pill medications in a medication organizer or pill box proves challenging, as these formats do not fit easily within the designated compartments.

The local medical center's outpatient care officer has raised concerns regarding the management of medications and dietary supplements in various formats during each medication refill visit. Specifically, the officer faces challenges when dealing with non-pill types of medications and dietary supplements, such as powders, syringes, and liquids. The unique characteristics of these non-pill formats cause difficulties for the officers in terms of storage, management, and dispensing. As a result, these officers encounter obstacles in ensuring accurate and timely provision of these medications and dietary supplements to older adults: *"The complex nature of handling and organizing non-pill formats demands additional attention and strategies to streamline the process and ensure optimal aging populations care."*

4.0.3 Fragmented information across multiple healthcare softwares. The current landscape of healthcare software, including healthcare portals, platforms, and various applications, presents a challenge as each system typically provides only partial or limited information. This fragmentation of information poses inconveniences for community-dwelling older adults who strive to comprehensively

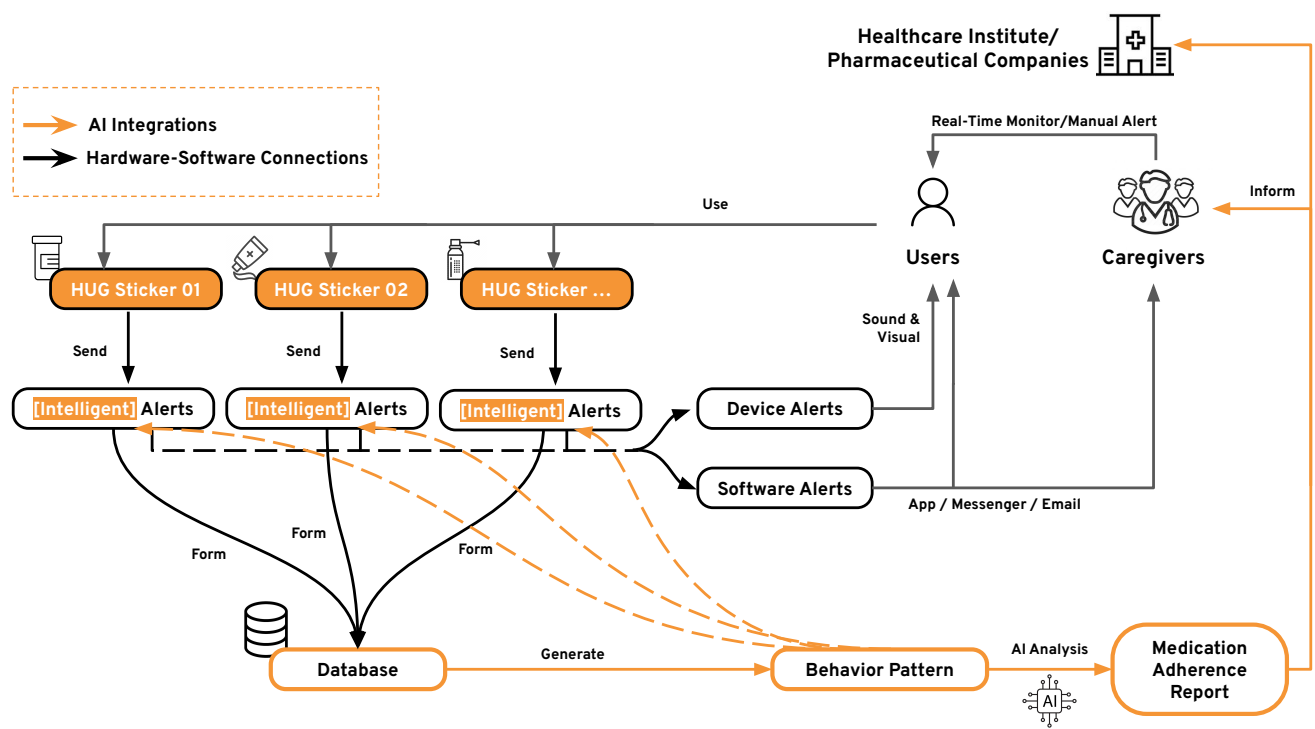


Figure 2: HUG Smart Sticker AIoT framework and workflow for managing multiple medications. (The interactive presentation of HUG Smart Sticker is available in the supplementary material video.)

Study 1 Caregiver ID	Department	Role
C1	Pharmacy	Consultant Pharmacist
C2	Clinical Integration	Clinical Preceptor
C3	Clinical Integration	Care Coordination Manager
C4	Nursing	Nurse

Table 1: Study 1 Caregivers Information

Study 1 Participant ID	Age	Gender	Education Level	Race	Family Income Level (estimate)
P1	73	F	Ph.D.	Caucasian	70k
P2	71	F	M.S.	Caucasian	30k
P3	73	F	M.S.	Caucasian	120k
P4	73	F	M.S.	Caucasian	80k
P5	71	F	M.S.	Caucasian	50k
P6	76	F	Ph.D.	Caucasian	60k
P7	68	M	Ph.D.	Caucasian	65k

Table 2: Study 1 Participants Information

understand their medications and dietary supplements systematically.

“As a patient I used several platforms, such as hospital’s patient platform, insurance system platform, and an application for my primary care provider. Specialists have different platforms, so there isn’t a single place I can go to get all my information. I do look up the

information from the platforms or applications, but I have to go to 3 different healthcare providers platforms to get a full picture of type of medications that I go through.” (P4)

The absence of a consolidated and holistic view of their healthcare data undermines the ability of older adults to attain a comprehensive understanding of their medications and dietary regimens. With fragmented information spread across multiple platforms, it becomes increasingly difficult for them to piece together a complete picture of their medications and dietary supplements. This lack of a centralized view often results in confusion, gaps in knowledge, and potential misconceptions.

5 Future Work

For future work, HUG Smart Sticker will conduct a usability testing study to prove its ability to improve the medication adherence rate within community-dwelling older adults. Rigorous usability testing will contribute invaluable data points, allowing us to iteratively optimize the HUG Smart Sticker’s design and functionality in response to user needs and preferences.

Secondly, building upon the usability testing study data points, HUG Smart Sticker would like to build AI predictions based on the aging users’ medication intake patterns and preferences, to deliver precise and target medication intake alerts. By harnessing AI capabilities, the HUG Smart Sticker endeavors to deliver highly tailored and timely medication intake alerts. This personalized approach not only aids caregivers in monitoring users effectively but also empowers individuals to gain profound insights into their medication behaviors.

Moreover, beyond age considerations, there’s a broader application for the HUG Smart Sticker in addressing medication adherence challenges associated with specific drug administration instructions. Some medications, such as thyroid medications, bisphosphonates, sucralfate, sildenafil, Captopril, Bethanechol, Ampicillin, and Zafirlukast, are recommended to be taken on an empty stomach [22]. By expanding the scope of the HUG Smart Sticker’s presentation to include a broader age range and diverse medication regimens, we can emphasize the technology’s versatility in addressing adherence challenges across different patient demographics.

6 Conclusions

In summary, the HUG Smart Sticker stands out as a unique and innovative intervention in comparison to conventional medication management methods. With its AIoT systematic approach and 3R features, this aging-friendly intervention allows community-dwelling older adults to seamlessly integrate medication and supplement management into their existing routines without the need for behavior modifications. The hardware module records usage data, while the software module provides reminders and generates personalized reports for users. This intervention proves particularly valuable for aging populations with mild cognitive impairments, enabling them to efficiently and effectively manage their medication and dietary supplement usage in real-time. In the long term, the widespread adoption of the HUG Smart Sticker could greatly assist millions of families with aging family members to intelligently manage medication usage. This AIoT system optimizes medication usage and minimizes the risk of medication MAD in

community-dwelling older adults living with mild cognitive impairment. This study will have a profound impact on the health and safety of community-dwelling older adults, empowering them to reduce medication non-adherence consequences and risks and continue to live at home while maintaining their independence and managing their health and well-being.

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A Appendix

B Task Analysis

We used comparative analysis based on the measurement adherence model [56] to show the comparisons between the existing medication management devices or systems and the HUG Smart Sticker (Table 3). By conducting a comparative analysis of the fourteen features implemented in the HUG Smart Sticker with those offered by four popular medication management services, it becomes evident that HUG Smart Sticker stands out by incorporating the 3R features (Record, Remind, and Report) and surpassing other services in key areas such as intelligent medication intake alerts, personalization features, and connections with caregivers. Especially for community-dwelling older adults, the HUG Smart Sticker accommodates users to maintain their original medication intake behavior, ensuring a sustainable and aging user-friendly medication management experience.



Figure 3: (Left) HUG Smart Sticker on three different medication packagings; (Middle) HUG Smart Sticker's hardware module; (Right) HUG Smart Sticker's software module.

	HUG Smart Sticker	Automatic Pill Dispenser	Monthly Pill Organizer System	Medacube	E-Pill Time Cap
Visual Alerts	✓	✗	✓	✓	✗
Sound Alerts	✓	✗	✗	✗	✗
Refill Status	✓	✗	✗	✓	✗
Size	< 10 cm	< 30 cm	< 20 cm	<30 cm	< 5 cm
Lock Feature	✗	✓	✓	✓	✓
Record Length (Per Cycle)	12 weeks	1.3 weeks	4 weeks	12.8 weeks	8 weeks
Battery	Battery	Battery	Battery	Charger	Battery
Traveller Friendly	✓	✗	✓	✗	✓
Aging Friendly Features	✓	✗	✓	✗	✗
Dosage Record	✓	✓	✓	✓	✗
Caregiver Connections	✓	✗	✗	✓	✗
Non-Solid Medication Management	✓	✗	✗	✗	✗
Software Monitor	✓	✗	✗	✗	✗
Hardware-Software Communications	Text, Email, etc	✗	✗	Text, Email, etc	✗

Table 3: HUG Smart Sticker feature task analysis with other four existing medication management devices

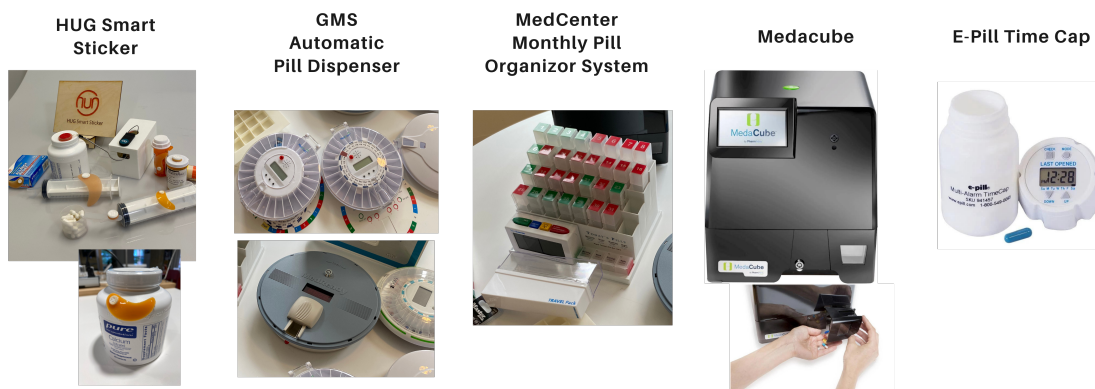


Figure 4: Tasks Analysis on existing medication management services